

SIMATS ENGINEERING
ASSESSMENT TOOL 3: Short Answer Test

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

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COURSE OUTCOME COVERED

CO1: Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions. (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1)
Question Count: 5

Total Marks: 50

Code of Conduct

I Shri Shanmuga Priya K 192324035 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Compare SOAP-based and REST-based web services in terms of communication style, data exchange, and suitability for cloud applications.
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage

Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Short Answer

1. Define Cloud Computing and Explain Any Four Characteristics that Make it Different from Traditional On-Premise Computing.

Definition

Cloud computing is a technology that delivers computing services such as servers, storage, databases, networking, software, and analytics over the Internet. Instead of purchasing and maintaining physical hardware, users can access these resources whenever required and pay only for what they use. Cloud computing provides flexibility, scalability, and cost efficiency for individuals and organizations.

Characteristics of Cloud Computing

1. On-Demand Self-Service

Cloud users can automatically provision computing resources such as virtual machines, storage, and applications whenever needed without requiring direct assistance from the service provider. This allows faster deployment and greater flexibility.

2. Broad Network Access

Cloud services are available over the Internet and can be accessed from different devices such as computers, laptops, tablets, and smartphones. Users can access their applications and data from anywhere with an Internet connection.

3. Resource Pooling

Cloud providers use a shared pool of computing resources to serve multiple customers. Resources such as servers, storage, and networks are dynamically allocated and reassigned according to customer demand while maintaining security and isolation.

4. Rapid Elasticity

Cloud computing allows resources to be increased or decreased quickly based on workload requirements. Organizations can easily scale their infrastructure during peak demand and reduce resources when demand decreases.

Difference from Traditional On-Premise Computing

Unlike traditional on-premise computing, where organizations purchase and maintain their own hardware and software, cloud computing offers resources over the Internet with minimal maintenance responsibilities. It reduces capital investment, improves scalability, and provides global accessibility.

Conclusion

Cloud computing offers a flexible and efficient way of delivering computing resources. Its characteristics such as on-demand service, network accessibility, resource sharing, and rapid scalability make it more effective than traditional computing systems.

2. Trace the Evolution of Cloud Computing from Hardware Evolution to Internet Software Evolution, and Show How These Changes Enabled Modern Cloud Platforms.

Introduction

Cloud computing did not emerge suddenly but evolved through several technological developments. Advancements in hardware, networking, virtualization, and Internet technologies gradually transformed the way computing resources are delivered.

Hardware Evolution

The earliest computers were large mainframe systems that were shared among multiple users through terminals. Later, personal computers became affordable, allowing individuals and organizations to own their own computing systems. This increased computing power but also required significant hardware investment and maintenance.

Client-Server Computing

The client-server architecture introduced dedicated servers that stored applications and data while client computers accessed these resources through a network. This improved resource sharing and centralized data management.

Internet Evolution

The rapid growth of the Internet connected computers worldwide and enabled users to access information and services remotely. Businesses started delivering services through websites and online applications, making remote access possible.

Virtualization

Virtualization technology allowed multiple virtual machines to run on a single physical server. This improved hardware utilization, reduced costs, and simplified infrastructure management. It became one of the most important technologies supporting cloud computing.

Internet Software Evolution

Software applications gradually moved from desktop installations to web-based applications that could be accessed through browsers. Services such as online email, document editing, and collaboration became common, encouraging the development of cloud-based software.

Modern Cloud Platforms

The combination of high-speed Internet, virtualization, distributed computing, and automated resource management led to the development of modern cloud platforms such as Amazon Web Services (AWS), Microsoft Azure, and Google Cloud Platform (GCP). These platforms provide scalable computing resources, storage, and software services on demand.

Conclusion

The evolution from hardware-based computing to Internet-based software and virtualization created the foundation for cloud computing. These technological advancements enabled the development of highly scalable, reliable, and cost-effective cloud platforms used around the world.

3. Explain the Role of Server Virtualization in Cloud Computing. Differentiate Between Virtualization as an Enabling Technology and Cloud Computing as a Service Model.

Introduction

Server virtualization is a technology that allows multiple virtual servers to operate on a single physical server. It improves resource utilization and forms the technological foundation of cloud computing.

Role of Server Virtualization

Server virtualization enables organizations to maximize the use of physical hardware by dividing a single server into multiple virtual machines. Each virtual machine operates independently with its own operating system and applications.

Virtualization increases hardware efficiency by allowing multiple workloads to share the same physical resources. It reduces the number of physical servers required, lowering costs for hardware, electricity, and maintenance.

It also provides flexibility by allowing virtual machines to be created, modified, migrated, or deleted quickly. This supports rapid deployment of applications and simplifies disaster recovery because virtual machines can be backed up and restored easily.

Server virtualization also improves security and reliability by isolating each virtual machine. A failure in one virtual machine does not directly affect the others.

Advantages of Server Virtualization

- Maximizes hardware utilization. Reduces infrastructure and maintenance costs.
- Supports rapid provisioning of virtual machines. Improves scalability and flexibility.
- Enhances disaster recovery and business continuity.
- Provides better isolation and security between applications.
- Simplifies server management and maintenance.

Virtualization and Cloud Computing

Virtualization is an enabling technology that creates virtual versions of computing resources, while cloud computing is a service model that delivers these virtualized resources over the Internet.

Virtualization focuses mainly on efficient hardware utilization within data centers. Cloud computing combines virtualization with automation, networking, and service management to provide computing resources to users on demand.

Although virtualization can exist without cloud computing, cloud computing depends heavily on virtualization to provide scalable and flexible services.

Conclusion

Server virtualization is the backbone of cloud computing because it improves resource utilization, reduces costs, and enables rapid scalability. Cloud computing extends virtualization by offering these resources as services over the Internet.

4. Compare SOAP-Based and REST-Based Web Services in Terms of Communication Style, Data Exchange, and Suitability for Cloud Applications.

Introduction

Web services enable communication between different applications over a network. SOAP and REST are two widely used approaches for implementing web services in cloud computing environments.

SOAP-Based Web Services

SOAP (Simple Object Access Protocol) is a protocol that uses XML for exchanging messages between applications. It follows strict communication standards and provides built-in security, reliability, and transaction support. SOAP services are commonly used in enterprise applications where security and reliability are critical.

REST-Based Web Services

REST (Representational State Transfer) is an architectural style that uses standard HTTP methods such as GET, POST, PUT, and DELETE. REST supports multiple data formats, including JSON and XML, although JSON is the most commonly used format because it is lightweight and faster.

Advantages of SOAP

- Provides strong security through WS-Security.
- Supports reliable message delivery.
- Offers transaction management.
- Platform and language independent.

Advantages of REST

- Lightweight and faster communication.
- Supports multiple data formats, especially JSON.
- Easy to develop and maintain.
- Stateless architecture improves scalability.

Comparison

SOAP relies entirely on XML messages and follows strict standards, making it more complex and slower than REST. REST uses standard web protocols and lightweight data formats, making it faster and easier to develop.

SOAP is suitable for applications requiring high security, reliable messaging, and formal contracts, such as banking and financial systems. REST is widely used in cloud computing because it provides better performance, lower bandwidth usage, and easier integration with web and mobile applications.

REST services are stateless, meaning each client request contains all the necessary information. This improves scalability, which is one of the major requirements of cloud applications.

Conclusion

Although SOAP provides strong security and reliability for enterprise systems, REST is generally preferred in cloud computing because of its simplicity, speed, scalability, and efficient communication.

5. Differentiate IaaS, PaaS, SaaS, and XaaS with One Suitable Real-World Example for Each Model.

Introduction

Cloud computing delivers services through different service models depending on the level of control and management required by users. The major service models are Infrastructure as a Service (IaaS), Platform as a Service (PaaS), Software as a Service (SaaS), and Anything as a Service (XaaS).

Infrastructure as a Service (IaaS)

Infrastructure as a Service provides virtualized computing resources such as servers, storage, networking, and virtual machines over the Internet. Users manage the operating system, applications, and data, while the cloud provider manages the physical infrastructure.

Example: Amazon Elastic Compute Cloud (Amazon EC2).

Platform as a Service (PaaS)

Platform as a Service provides a complete environment for developing, testing, and deploying applications. The provider manages the hardware, operating system, and development platform, allowing developers to focus only on application development.

Example: Google App Engine.

Software as a Service (SaaS)

Software as a Service delivers fully developed software applications over the Internet. Users simply access the application through a web browser without worrying about installation, maintenance, or updates.

Example: Microsoft Office 365 or Gmail.

Anything as a Service (XaaS)

Anything as a Service is a broad term that includes all cloud-based services beyond IaaS, PaaS, and SaaS. It covers services such as Database as a Service (DBaaS), Security as a Service (SECaaS), Storage as a Service (STaaS), and Artificial Intelligence as a Service (AIaaS).

Example: MongoDB Atlas (Database as a Service).

Conclusion

Each cloud service model provides a different level of responsibility and management. IaaS offers infrastructure, PaaS provides a development environment, SaaS delivers complete software applications, and XaaS extends cloud technology to almost every type of IT service. These service models enable organizations to choose the most suitable solution based on their requirements.